Boundary Conformal Field Theories

Samuel Homiller

University of Illinois at Urbana-Champaign: Physics 583

April 27, 2015

The goal of this paper is to introduce some of the interesting aspects of conformal field theories (CFTs) on manifolds with a boundary. Conformal invariance on the infinite plane yields a number of powerful results — particularly in applications to statisitcal mechanical systems and string theories. Boundary conformal field theories are in a sense more physical than their idealized infinite counterparts, but their study has also led to a number of more abstract results as well, which we'll attempt to describe below.

This paper is organized into three main sections. In the first we review the basics of conformal field theory on the infinite $2d$ plane and its quantization and introduce the minimal models and the concept of modular invariance. In the second we introduce boundary CFTs, describe how to calculate correlation functions on a manifold with boundary, and introduce the notion of a boundary operator. In the final section we mention some applications of CFTs with boundary in string theories and discuss its application to the critical percolation problem.

1 Introduction - Conformal Field Theory

While ideas of general conformal invariance have been understood for some time, the real history of the subject began with the work of Belavin, Polyakov and Zamolodchikov (BPZ) who showed that the conformal algebra becomes infinite dimensional in two dimensions [1]. For this reason, two dimensional CFT will be the primary focus in this paper. All the concepts introduced in this section are flushed out in greater detail in [2], the standard reference on CFTs. Other useful introductions can be found in [3, 4, 5, 6, 7].

1.1 2-dimensional CFTs

We work in two Euclidean dimensions. Under a change of coordinates $z \to w(z)$, the metric transforms as

$$g^{\mu\nu} \to \left(\frac{\partial w^\mu}{\partial z^\alpha}\right)\left(\frac{\partial w^\nu}{\partial z^\beta}\right) g^{\alpha\beta} \tag{1}$$

By definition, this transformation is conformal if the transformed metric is invariant up to a scale, $g'(x') = \Lambda(x)g(x)$. In two dimensions, this implies:

$$\frac{\partial w^0}{\partial z^0}\frac{\partial w^1}{\partial z^0} + \frac{\partial w^0}{\partial z^1}\frac{\partial w^1}{\partial z^1} = 0 \tag{2}$$

Locally, this condition is clearly satisfied if $w(z)$ satisfies the Cauchy-Riemann equations for holomorphic or anti-holomorphic functions. It is thus beneficial to extend z^0 and z^1 to the complex plane, and use (independent) coordinates $z = z^0 + iz^1$, $\bar{z} = z^0 - iz^1$.[1] In this language, we see that any mapping of the

[1] As discussed in [2], extending z^0 and z^1 to $\mathbb{C}$ allows use to regard z and $\bar{z}$ as independent variables. One should note, however, that the physical surface is defined by $\bar{z} = z^*$

form $z \to w(z)$ is conformal if $w(z)$ is analytic (the anti-holomorphic coordinate $\bar{z}$ transforms likewise under any anti-holormorphic map, $\bar{w}$). In a conformally invariant theory, the energy-momentum tensor $T_{\mu\nu}$ is traceless. In terms of complex coordinates, this implies that $T_{z\bar{z}} = T_{\bar{z}z} = 0$, and $\partial_{\bar{z}} T_{zz} = \partial_z T_{\bar{z}\bar{z}} = 0$. So the two non-zero components are completely holomorphic and anti-holomoprhic, respectively, and we can write $-2\pi T_{zz} = T(z)$ and $-2\pi T_{\bar{z}\bar{z}} = \bar{T}(\bar{z})$ (the -2π is for convenience).

While we are usually interested in local conformal invariance, we must sometimes consider globally conformal mappings. These take the special form

$$w(z) = \frac{az+b}{cz+d}, \quad ad - bc = 1 \qquad a, b, c, d \in \mathbb{C} \tag{3}$$

and form the group $SL(2, \mathbb{C})$, of linear fractional transformations.

A field $\phi(z, \bar{z})$ is called a primary field if under any local conformal transformation, the variation of ϕ is

$$\delta_{\epsilon,\bar{\epsilon}}\phi = -(h\phi\partial_z\epsilon + \epsilon\partial_z\phi) - (\bar{h}\phi\partial_{\bar{z}}\bar{\epsilon} + \bar{\epsilon}\partial_{\bar{z}}\phi) \tag{4}$$

where the quantities h, $\bar{h}$ are the holomorphic and anti-holomorphic conformal dimensions respectively. If the field ϕ has scaling dimension Δ and planar spin s, the conformal dimensions are given by $h = \frac{1}{2}(\Delta + s)$ and $\bar{h} = \frac{1}{2}(\Delta - s)$. A field is quasi-primary if under a conformal map $z \to w(z)$ it transforms as

$$\phi'(w, \bar{w}) = \left(\frac{dw}{dz}\right)^{-h} \left(\frac{d\bar{w}}{d\bar{z}}\right)^{-\bar{h}} \phi(z, \bar{z}). \tag{5}$$

Conformal invariance, even in higher dimensions, also implies a number of constraints on the form of correlation functions that we will on occassion take for granted without discussion (see [2] for details).

As with any symmetry transformation, conformal invariance implies a Ward identity with regards to correlation functions of quasi-primary fields. This takes the form

$$\delta_{\epsilon,\bar{\epsilon}} \langle X \rangle = -\frac{1}{2\pi i} \oint_C dz \epsilon(z) \langle T(z) X \rangle + \frac{1}{2\pi i} \oint_C d\bar{z} \bar{\epsilon}(\bar{z}) \left\langle \bar{T}(\bar{z}) X \right\rangle \tag{6}$$

where X stands for a product of a finite number of primary fields, $\phi(z_1, \bar{z}_1) \ldots \phi(z_n, \bar{z}_n)$. This relation is known as the conformal Ward Identity.

Given two fields $A(z)$ and $B(w)$, we can express the correlation function of A and B in a series called the operator product expansion (OPE):

$$A(z)B(w) = \sum_{n=-\infty}^{N} \frac{\{AB\}_n(w)}{(z-w)^n} \tag{7}$$

where the $\{AB\}_n$ are fields that are nonsingular at $z = w$. The OPE is valid only within correlation functions, but we generally omit the brackets. Examples of the OPE for various free fields are given in [2]. Of particular importance is the OPE of the energy-momentum tensor of a theory with itself[2]

$$T(z)T(w) \sim \frac{c/2}{(z-w)^4} + \frac{2T(w)}{(z-w)^2} + \frac{\partial T(w)}{(z-w)} \tag{8}$$

where the constant c is the central charge of the theory, which gives a rough indication of the short-distance behavior of the theory (see [2] for an illuminating discussion). For the free boson, fermion, and Fadeev-Popov ghost systems, the central charges are 1, $\frac{1}{2}$, and -26 respectively. One of the most important advancements in BPZ [1] was recognizing the central charge as a fundamental characteristic of any conformal field theory.

[2]Following [2], we use the symbol $\sim$ in OPEs to mean equality modulo terms that are regular as $w \to z$.

1.2 The Virasoro Algebra

To make full use of the analytic structure of conformal field theories in $2d$, it is most straightforward to use the so-called operator formalism by means of radial quantization. Beginning with space and time coordinates x and t, with x compactified to a length L, we use the map $z = \exp(2\pi(t+ix)/L)$. Under this mapping, time runs along the radial direction, with $t = \pm\infty$ at $|z| = \infty, 0$ respectively. The spatial coordinate x then runs along concentric circles about the origin. We build a Hilbert space by assuming a vacuum state[3] $|0\rangle$, and define fields as acting on this vacuum state, $|\phi_{in}\rangle = \lim_{z,\bar{z}\to 0} \phi(z,\bar{z})\,|0\rangle$.[4] Now, we can write mode-expansions for fields:

$$\phi(z,\bar{z}) = \sum_{m\in\mathbb{Z}} \sum_{n\in\mathbb{Z}} z^{-m-h} \bar{z}^{-n-\bar{h}} \phi_{m,n} \tag{9}$$

$$\phi_{m,n} = \frac{1}{2\pi i} \oint dz\; z^{m+h-1} \frac{1}{2\pi i} \oint d\bar{z}\; \bar{z}^{n+\bar{h}-1} \phi(z,\bar{z}). \tag{10}$$

This method of contour integration admits a practical relation between commutators and the operator product expansion, using the residue theorem within the integration. We can expand the energy-momentum tensor in the same way:

$$T(z) = \sum_{n\in\mathbb{Z}} z^{-n-2} L_n; \qquad L_n = \frac{1}{2\pi i} \oint dz\; z^{n+1} T(z). \tag{11}$$

The L_n, and $\bar{L}_n$ are the generators of conformal transformations on the Hilbert space. Using the OPE of $T(z)$ and the relationship between commutators and contour integration in radial quantization, we can derive the algebra satisfied by the L_n, $\bar{L}_n$:

$$[L_n, L_m] = (n-m)L_{n+m} + \frac{c}{12} n(n^2-1)\delta_{n+m,0} \tag{12}$$

$$\left[L_n, \bar{L}_m\right] = 0 \tag{13}$$

$$\left[\bar{L}_n, \bar{L}_m\right] = (n-m)\bar{L}_{n+m} + \frac{c}{12} n(n^2-1)\delta_{n+m,0} \tag{14}$$

these operators and their commutation relations form the Virasoro algebra. Now the asymptotic primary fields are eigenstates of L_0 and $\bar{L}_0$. We write $|h,\bar{h}\rangle = \phi(0,0)\,|0\rangle$, so that $L_0\,|h,\bar{h}\rangle = h\,|h,\bar{h}\rangle$. The generators L_n with $n < 0$ increase the dimension of these eigenstates, so we can form "descendants" of the state $|h\rangle$ by repeatedly applying these generators.

1.3 Minimal Models

Analogous to what is done for the spin algebra, conformally invariant theories are often built with the use of highest-weight representations. We denote a highest-weight state, $|h\rangle$ (an eigenstate of L_0 with eigenvalue h), and then form descendant states as described above. The representation space formed around the generator L_0 is called a Verma module, and is usually labeled $V(c,h)$. The character of the module is defined as

$$\chi_{(c,h)}(\tau) = \mathrm{tr} q^{L_0 - c/24} \qquad (q = e^{2\pi i \tau}) \tag{15}$$

$$= \sum_{n=0}^{\infty} \dim(h+n) q^{n+h-c/24} \tag{16}$$

where $\dim(h+n)$ is the number of linearly independent states at level n in the module. The argument τ will appear in Section 1.4 as the ratio of the two periods of a torus. It can be shown [2] that

$$\chi_{(c,h)}(\tau) = \frac{q^{h+(1-c)/24}}{\eta(\tau)} \tag{17}$$

[3] For interacting theories, we attenuate the interaction at $t = \pm\infty$.

[4] The conjugate field is defined as $[\phi(z,\bar{z})]^\dagger = \bar{z}^{-2\bar{h}} z^{-2h} \phi(1/z, 1/\bar{z})$, so that the Hermitian product has the required form of a conformally invariant two-point function.

where η is the Dedekind function. Generally, we will consider unitary representations of the Virasoro algebra, which must satisfy certain constraints on h, and c. Unitary, reducible, theories are in correspondence with roots of the Kac determinant [2]. One of the greatest triumphs of early conformal field theory, led by Belavin, Polyakov and Zamolodchikov [1] and Friedan, Qiu and Shenker [8], was the discovery and classification of the minimal models — unitary theories with a finite number of primary operators. Following [1], we parameterize c as

$$c = 1 - \frac{6}{m(m+1)} \tag{18}$$

for rational m. The finite set of operators now contains only fields with scaling dimensions given by the Kac determinant:

$$h_{p,q} = \frac{(p(m+1) - qm)^2 - 1}{4m(m+1)}. \tag{19}$$

The minimal models are significant not only because they can be described in detail mathematically, but also because they correspond precisely to different statistical systems (see [2] for a summary). In general, these models are found by identifying the null states in a Verma module of the form described above. The vanishing of correlation functions of null states and other fields implies differntial equaitons that must be satisfied by general correlation functions. These relations help establish the fusion rules of the theory — the conditions under which a fields in a certain conformal family appear in the product of two conformal fields. Minimal models are theories where this derived operator algebra truncates at a certain level (see [2] for many more details). In general, we define the fusion number, ${\mathcal{N}_{ij}}^k$, as the number of times the field ϕ_k appears in the operator product expansion of ϕ_i and ϕ_j.[5].

1.4 Modular Invariance and the Verlinde Formula

Thus far we've considered models to be a combination of the decoupled holomorphic and anti-holomoprhic theories, which could in principle have been different. However this decoupling is not very physical. The most simple connections between the two parts of a theory arise from demanding that the partition function of the whole theory be invariant under modular transformations.

Modular invariance arises when considering a conformal field theory on a torus — a two-dimensional surface compactified along two periods, ω_1, and $\omega_2 \in \mathbb{C}$. The ratio of these two periods is the modular parameter mentioned above: $\tau = \omega_2/\omega_1$. For a minimal theory, the partition function of the theory on the torus is given by

$$Z(\tau) = \sum_{(r,s),(t,u)\in E_{p,p'}} \mathcal{M}_{rs;tu}\chi_{r,s}(\tau)\bar{\chi}_{t,u}(\bar{\tau}) \tag{20}$$

where the indices parameterize the central charge and dimension, limited to the sets of pairs corresponding to the minimal models (see [2] for details). The factor $\mathcal{M}_{rs;tu}$ is the multiplicity of the particular module in the theory. Any modular transformation can be built from a combination of

$$\mathcal{T} : \tau \to \tau + 1 \qquad \mathcal{S} : \tau \to \frac{-1}{\tau}. \tag{21}$$

Modular invariant theories thus satisfy $Z(\tau+1) = Z(\tau)$ and $Z(-1/\tau) = Z(\tau)$. The simplest way to satisfy this invariance is to simply set $h = \bar{h}$ in each pairing of holomorphic and anti-holomorphic theories (the diagonal models). However, there are more complicated theories, described in some detail in [2].

Considering, in particular, the transformation $\mathcal{S}$ on "conformal blocks" of the theory (see [2]), it can be demonstrated that $\mathcal{S}$ diagonalizes the fusion numbers in these conformal blocks. This leads to the Verlinde formula for the fusion numbers in terms of the matrix elements of $\mathcal{S}$ in the basis of conformal characters:

$${\mathcal{N}_{ij}}^k = \sum_m \frac{\mathcal{S}_{im}\mathcal{S}_{jm}\bar{\mathcal{S}}_{mk}}{\mathcal{S}_{0m}}. \tag{22}$$

[5] In the Virasoro minimal models, the fusion numbers are always 0 or 1, but this is not necessarily true in more general theories[2]

2 Conformal Invariance on Surfaces with Boundary

This section will apply the ideas of conformal invariance described above to surfaces with boundaries, where boundary conditions have to be taken into account. This subject was largely pioneered single-handedly by John Cardy in a series of papers in the 1980s [9, 10, 11]. Cardy demonstrated that boundary conditions constrained the form of correlation functions of conformally invariant theories restricted to the upper-half plane and provided a means of determining the form of these correlators using a "method of images", which he applied to several of the common minimal models in [9]. Later, Cardy showed that if the boundary conditions were different on different regions of the boundary, so-called "boundary operators" would arise [11]. Cardy demonstrated that these boundary operators provide a clear relationship between the operator content of a theory on an annulus and its fusion rules. Each of these results is discussed in [2], and are described in more detail in the following sections.

2.1 The Upper Half Plane

Following [9], we begin with the upper half place (that is, $z \in \mathbb{C} : \mathrm{Im}(z) > 0$) because it is the simplest manifold in two-dimensions with a boundary. This does not limit the discussion, however, as many manifolds with boundary can be obtained via conformal transformations on the upper half-plane (see section 2.2). In order to maintain the maximum amount of conformal symmetry, a conformal transformation must map the boundary to itself and preserve the boundary conditions of the model. This restriction will couple the holomorphic and anti-holomorphic sectors of the theory in interesting ways.

For global conformal transformations of the form 3, it is easy to see that the real axis is preserved if the parameters a, b, and c are real.[6] Locally, the real axis is preserved under a transformation $z \to z + \epsilon(z)$ if ϵ is real on the real axis (i.e., $\epsilon(\bar{z}) = \bar{\epsilon}(z)$). While the conformal Ward identity is still valid for contours restricted in the upper half-plane, it is easiest to follow Cardy and define $T(z) = \bar{T}(z)$ for $\mathrm{Im}(z) < 0$, and consider the contour $\bar{C}$ as shown in Figure 1. The conformal Ward identity now reads:

$$\delta_{\epsilon,\bar{\epsilon}} \langle X \rangle = -\frac{1}{2\pi i} \oint_C dz \epsilon(z) \langle T(z) X \rangle + \frac{1}{2\pi i} \oint d\bar{z} \bar{\epsilon}(\bar{z}) \left\langle \bar{T}(\bar{z}) X \right\rangle \tag{23}$$

$$= -\frac{1}{2\pi i} \oint_C dz \epsilon(z) \langle T(z) X \rangle + \frac{1}{2\pi i} \oint_{\bar{C}} dz \epsilon(z) \langle T(z) X^* \rangle \tag{24}$$

where in the second line we've taken the conjugate of all points z and integrated over the mirrored contour, $\bar{C}$ and X and X^* now represent only the holomorphic parts of the fields, evaluated at z and z^* respectively. As we take the horizontal parts of the contours to be infinitesimally close to the real axis, the two contributions cancel, since $T = \bar{T}$. We can now combine these two contours into one, enveloping the holomorphic coordinates z and their complex conjugates.

The virtue of this method is that we have now replaced all the degrees of freedom in the anti-holomorphic part of the fields with the holomorphic parts of the fields, evaluated at their complex conjugates. Thus, the correlator of primary fields on the upper half-plane as a function of the variables $z_1, \ldots z_n$ and their anti-holomorphic counterparts must solve the same differential equation as the correlator of the fields at the $2n$ holomorphic coordinates (with $z_{i+n} = z_i^*$) on the full complex plane without boundary.

2.2 The Infinite Strip

As an example of how the upper half plane can be mapped onto more general surfaces with boundary, we consider the spin-spin correlation function in the Ising model, following [2]. The form of the general two-point

[6] Note that because we've fixed $ad - bc = 1$, this implies that d is real as well.

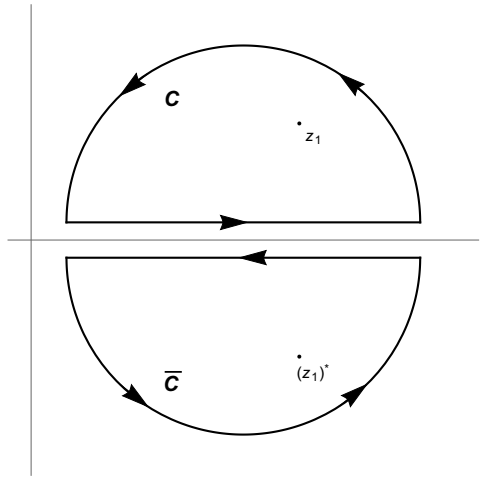

Figure 1: Contours C and $\bar{C}$ used for the conformal Ward identity on the upper-half plane.

correlator on a strip was first calculated by Cardy [12]. The relevant mapping is

$$w(z) = \frac{L}{\pi} \log z. \tag{25}$$

This maps the entire complex plane onto a strip of width $2L$, with periodic boundary conditions (a cylinder). However, when z is restricted to the upper half plane, the map is onto the strip $\mathrm{Im}(w) \leq L/2$, where the boundary conditions both sides of the strip match the boundary conditions on the real axis — the positive real axis is mapped onto the lower edge of the strip, and the negative real axis onto the upper edge. Now, in the infinite plane, the spin-spin correlator has the form:

$$\langle \sigma(z_1, \bar{z}_1)\sigma(z_2, \bar{z}_2)\rangle \sim (z_1 - z_2)^{-2h}(\bar{z}_1 - \bar{z}_2)^{-2\bar{h}}. \tag{26}$$

After mapping onto the half-plane, we instead get the form of a four-point correlator of only holomorphic variables, $\langle \sigma(z_1)\sigma(z_2)\sigma(z_1^*)\sigma(z_2^*)\rangle$. Using the map (25), we find the general form

$$\langle \sigma(w_1, \bar{w}_1)\sigma(w_2, \bar{w}_2)\rangle = \left(\frac{\pi}{L}\right)^{\frac{1}{4}} \left[e^{2piu_1/L} e^{piu/L}\right]^{\frac{1}{8}} \langle \sigma(z_1)\sigma(z_2)\sigma(z_1^*)\sigma(z_2^*)\rangle \tag{27}$$

where $w = u + iv$. The bulk four-point correlator is given by (see [2] for details)

$$\langle \sigma\sigma\sigma\sigma\rangle = \left(\frac{z_{13}z_{24}}{z_{12}z_{23}z_{14}z_{34}}\right)^{\frac{1}{8}} F(x) \tag{28}$$

where $z_{ij} = z_i - z_j$ and x is the anharmonic ratio:

$$x = \frac{z_{12}z_{34}}{z_{13}z_{24}}. \tag{29}$$

Now, in the limit of long distances along the strip, $u_1 - u_2 \to \infty$, $x \sim e^{\pi(u_1-u_2)/L}$ and we find

$$\langle \sigma(w_1, \bar{w}_1)\sigma(w_2, \bar{w}_2)\rangle \sim e^{-\pi(u_1-u_2)/2L}. \tag{30}$$

So, the spin correlation function has an exponential decay, with correlation length $\xi = 2L/\pi$. This result was derived by Cardy [12], who matched it to existing numerical calculations.

2.3 Boundary Operators

In the previous section, we obtained the finite-width strip geometry via a conformal map from the upper-half plane, leading to the same conformal boundary condition (which we left unspecified) on each side of the strip. However, conformal invariance does not *a priori* demand that the boundary condition on each edge be the same. If we instead started with the infinite strip, with different boundary conditions (labelled α and β for later use) on each edge and used the inverse (exponential) map to the upper-half plane, we would find a discontinuity in the boundary conditions at $z = 0$. Cardy proposed that this discontinuity was equivalent to inserting a "boundary condition changing operator" at $z = 0$ [11]. Considerations of such operators lead to surprising conclusions about the fusion rules of the theory and an alternate realization of the Verlinde formula that we'll summarize below.

Following [2], boundary operators can easily be seen to emerge via the method of images described above: as we move a bulk field $\phi(z)$ on the upper-half plane infinitesimally close to the boundary, it interacts with the mirror field $\phi(z^*)$, and can be replaced by the OPE of the two:

$$\phi(z)\phi(z^*) \sim \sum_i (z - z^*)^{(h_i - 2h)} \phi_B^{(i)}((z + z^*)/2) \tag{31}$$

The correspondence between these two ideas is demonstrated via two equivalent methods of quantizing on a finite cylinder: where time flows around and along the cylinder respectively. If we wrap the finite strip described above onto a cyliner, we get different boundary conditions α and β on the two edges. The Hamiltonian with time moving around the cylinder thus depends on the boundary conditions at the edges, and we denote it $(\pi/L)H_{\alpha\beta}$. We can thus write a partition function,

$$Z_{\alpha\beta} \quad = \quad \mathrm{tr} q^{H_{\alpha\beta}} = \sum_i n^i_{\alpha\beta} \chi_i(q) \tag{32}$$

where q is the same factor as before, with $\tau = iT/2L$ (for a cylinder of circumference T and length L). In the second line, we've written out the trace, with χ_i the character of the Virasoro representation i and $n^i_{\alpha\beta}$ the number of copies of that representation in the spectrum. Now, under the mapping $\mathcal{S}$ defined in Section 1.4, equation (32) still holds, but with χ_i replaced by $\sum_j \mathcal{S}_{ij}\chi_j(e^{-2\pi i/\tau})$. In this context though, time moves along the length of the cylinder, and the boundary conditions are realized as initial and final states, $|\alpha\rangle$, $|\beta\rangle$. Now, mapping onto the ξ-plane, H takes the form

$$\tilde{H} = \frac{2\pi}{T}\left(L_0^\xi + \bar{L}_0^\xi - \frac{c}{12}\right) \tag{33}$$

While we have not specified the particular boundary conditions, α and β, we can generally demand that no energy flow across the edge of the cylinder. In the language of the ξ-plane, this requires

$$\left(L_n^\xi - \bar{L}_n^\xi\right)|\alpha\rangle = 0 \tag{34}$$

and similar for β. Now, taking all the states $|j\rangle$ (regardless of which representation they belong to) that satisfy (34), we can express $|\alpha\rangle$ and $|\beta\rangle$ as linear combinations and write the partition function in the ξ plane as

$$Z_{\alpha\beta}(q) = \sum_j \langle\alpha|j\rangle \langle j|\beta\rangle \xi_j(e^{-2\pi i/\tau}) \tag{35}$$

Comparing this to the original formulation of the partition function, we find

$$\sum_i \mathcal{S}_{ij} n^i_{\alpha\beta} = \langle\alpha|j\rangle \langle j|\beta\rangle \tag{36}$$

In a unitary model, we can find a boundary state that only appears in the identity representation, $|\tilde{0}\rangle$, and write $|\tilde{0}\rangle = \sum_j \sqrt{\mathcal{S}_{0j}}\,|j\rangle$. Then an arbitrary boundary state $|\ell\rangle$ can be written

$$|\ell\rangle = \sum_j \frac{\mathcal{S}_{\ell j}}{\sqrt{\mathcal{S}_{0j}}}\,|j\rangle \tag{37}$$

But with this replacement, Eq. 36 becomes

$$\sum_i \mathcal{S}_{ij} n^i_{\alpha\beta} = \frac{\mathcal{S}_{\alpha i}\mathcal{S}_{\beta j}}{\mathcal{S}_{0j}} \tag{38}$$

which is equivalent to the Verlinde formula. The appearance of the Verlinde formula here solidifies our interpretation of the boundary states as boundary condition changing operators, and we identify $n^i_{k\ell} = \mathcal{N}^i_{kl}$. The insertion of a local operator confined to the boundary indeed changes the boundary conditions by allowing states from different representations of the algebra propogate via the non-zero fusion coefficients.

3 Applications of Boundary CFTs

Conformal invariance on manifolds with a boundary has made stronger the connection between minimal theories, statiscal models, and physical systems. However, it has a surprising number of applications outside of statistical mechanics as well, particularly in string theory. This connection will be qualitatively discussed below, and another application — in the problem of critical percolation — will be flushed out in some detail.

3.1 Conformal Invariance in String Theory

Conformal field theory is one of the most powerful mathematical tools in string theory, where the action can be written along the two-dimensional string worldhseet. Conformal invariance is realized as a freedom in choosing the worldsheet metric, as we'll derive below. The main reference for this section was the text by Polchinski [13], whose notation will be followed closely. We start with the Euclidean Polyakov action:

$$S = S_X + S_\chi \tag{39}$$

$$S_X = \frac{1}{4\pi\alpha'}\int_M d^2\sigma g^{1/2} g^{ab}\partial_a X^\mu \partial_b X_\mu \tag{40}$$

$$S_\chi = \frac{1}{4\pi}\int_M d^2\sigma g^{1/2} R + \frac{1}{2\pi}\int_{\partial M} dsk \tag{41}$$

where R is the Ricci scalar, k is the geodesic curvature, g_{ab} is the worldsheet metric (which runs over σ^1, σ^2), and the X^μ are the positions of the worldsheet in d-dimensional spacetime, parameterized over the (Euclidean) worldsheet coordinates, σ^i. In addition to Poincar'e invariance, this action is also invariant under a change of coordinates $(X'^\mu(\sigma^{1'}, \sigma^{2'}) \to X^\mu(\sigma^1, \sigma^2))$ – often referred to as diffeomorphism invariance – and so-called Weyl invariance: a local rescaling of the world-sheet metric.

As in gauge theories, naively performing the path integral over all possible configurations $[X, g]$, would be drastically overcounting due to the symmetries described above. Thus, a gauge fixing procedure is needed. Nominally, there exists enough gauge freedom to eliminate the integration over the metric entirely, but for reasons we won't discuss, it is frequently convenient to work in conformal gauge:

$$\hat{g}_{ab}(\sigma) = \exp\left(2\omega(\sigma)\right)\delta_{ab} \tag{42}$$

where $\omega(\sigma)$ is an arbitrary function of the worldsheet coordinates. Again, as in gauge theories, this process of gauge fixing introduces ghosts into the path integral. For conformal gauge, the ghost action is

$$S_g = \frac{1}{2\pi}\int d^2z \left(b_{zz}\bar{\partial}c^z + b_{\bar{z}\bar{z}}\partial c^{\bar{z}}\right) \tag{43}$$

where we've switched to holomorphic and anti-holomorphic coordinates, z, $\bar{z}$. This ghost action defines a particular conformal field theory, and as is shown in [2], the central charge of this theory alone is $c_g = -26$. Now, we require that the action be Weyl-invariant not only classically, but at the quantum level as well. This

requires cancellation of the so-called Weyl anomaly: $T^a_a \propto R$. It is shown in [13] that this proportionality factor is related to the central charge of the theory, and thus Weyl invariance requires the central charge of the theory to vanish. This vanishing occurs only if the central charge of the D scalar components, X^μ, is opposite that of the ghosts. Since a scalar theory has $c = 1$, we see that $D = 26$, a well known feature of bosonic string theory.

In the above we have implicitly assumed that the string is closed — that is, the ends of the string connect to form a loop, so the worldsheet is topologically a cylinder. However, we can also consider open strings, which then look closer to a finite width strip. Quantizing such a string worldsheet naturally requires boundary conditions to be taken into account. Polchinksi realized that choosing Dirichlet-boundary conditions on the edges led to new dynamical objects he called Dirichlet-branes (D-branes) [14]. Since then, there has been a great deal of activity surrounding boundary conditions in string theory. This approach to string theory is summarized nicely in [4], where some more recent results are discussed as well.

3.2 Critical Percolation and the Q-state Potts Model

Perhaps surprisingly, boundary operators in conformal field theory yield a powerful application in the study of critical percolation. Less surprisingly, this application was also worked out by Cardy [15]. A brief review of these methods is given in [2], the key results of which we summarize here.

Bond percolation is the study of a finite rectangular lattice, where sites are connected by bonds that can be turned either on or off, with probability p. The goal is to determine the probability $\pi(p)$, that there is a path of only activated bonds from the left to right side of the lattice. In the limit of infinite lattice size, there is a critical value p_c, below which $\pi(p)$ vanishes, and above which $\pi(p)$ goes to 1. We wish to determine the value of p_c as a function of the aspect ratio, r, of the rectangular lattice.

The relationship to conformal field theory arises from modeling this as a Q-state Potts model, which corresponds to a class of minimal models. A Q-state Potts model is simply a lattice model where an energy J arises from nearest neighbor sites having the same value of spin, and energy 0 otherwise. In terms of the percolation lattice, we let B be the number of bonds on the lattice, R the subset of activated bonds, $B(R)$ the number of activated bonds, and $N_c(R)$ the number of disjoint clusters in the subset R so that we can write the Q-state Potts model partition function as [7]

$$Z = \sum_R p^{B(R)}(1-p)^{B-B(R)} Q^{N_c(R)} \tag{44}$$

Now, we write $Z_{\alpha\beta}$, the partition function on the lattice with the spins fixed in state α on the left and β on the right, but with free boundary conditions on the top and bottom. The crossing probability is then given by $\pi(r) = \lim_{Q\to 1}(Z_{\alpha\alpha} - Z_{\alpha\beta})$, where now:

$$Z_{\alpha\alpha} - Z_{\alpha\beta} = \sum_R{}' p^{B(R)}(1-p)^{B-B(R)} \tag{45}$$

Now, if $x_1, \ldots x_4$ are the coordinates of the four corners of the lattice, the partition functions take the form

$$Z_{\alpha\alpha} = Z_f \langle \phi_{fa}(x_1)\phi_{\alpha f}(x_2)\phi_{f\alpha}(x_3)\phi_{\alpha f}(x_4)\rangle \tag{46}$$

$$Z_{\alpha\beta} = Z_f \langle \phi_{fa}(x_1)\phi_{\alpha f}(x_2)\phi_{f\beta}(x_3)\phi_{\beta f}(x_4)\rangle \tag{47}$$

where the ϕ are boundary condition changing operators (as described in Section 2.3) and Z_f is the partition function for free boundary conditions (f). The Q-state Potts model with $Q = 1$ corresponds to the minimal

[7] In order that our later definitions be well defined, we let Q be arbitrary for now, and we'll take the correct limit $Q \to 1$ later. Note that Z is normalized correctly if $Q = 1$.

model $\mathcal{M}(3,2)$, with central charge $c = 0$ [2]. With this identification, we can fix the form of each of the four fields. Now, we conformally map the boundary of the rectangle onto the real axis using a Schwarz-Christoffel transformation, and it remains to find the form of $\langle\phi(z_1)\phi(z_2)\phi(z_3)\phi(z_4)\rangle$, where the ϕ are of a particular functional form that arises in the Ising model. From here, the form of the differential equation that this correlation function satisfies can be found: its solutions are related to the hypergeometric function, F. Now, defining the Schwarz-Chirstoffel transformation appropriately to take account of the aspect ratio, r, we eventually find that the crossing probability is

$$\pi(r) = \frac{3\Gamma(\frac{2}{3})}{\Gamma(\frac{1}{3})^2} x^{1/3} F(\frac{1}{3}, \frac{2}{3}, \frac{4}{3}; x), \tag{48}$$

a result that is in fantastic agreement with numerical simulations (see [2]).

In conclusion, consideration of boundary conditions in conformally invariant quantum field theories has led to a number of exciting results and applications of conformal field theory. Imposing bounary conditions makes closer the connection of statistical models and physical $2d$ systems and has led to accurate predictions within these theories. Boundary CFTs also tighten the relationship between conformal invariance and string theory by allowing consideration of open strings. Finally, their application in calculating critical percolation remains one of the most accurate numerical predictions of conformal invariance to date.

References

[1] A. Belavin, A. Polyakov, and A. Zamolodchikov, Nucl. Phys. B **241**, 333 (1984).

[2] P. D. Francesco, P. Mathieu, and D. Senechal, *Conformal Field Theory* (Springer, 1997).

[3] P. Ginsparg, Fields, Strings and Critical Phenomena, (Les Houches, Session XLIX, 1988) (1988), arXiv:hep-th/9108.028.

[4] A. Recknagel and V. Schomerus, *Boundary Conformal Field Theory and the Worldsheet Approach to D-Branes* (Cambridge University Press, 2014).

[5] M. R. Gaberdiel, (1999), arXiv:hep-th/9910.156.

[6] R. Blumenhagen and E. Plauschinn, *Introduction to Conformal Field Theory with Applications to String Theory* (Springer, 2009).

[7] M. Schottenloher, *A Mathematical Introduction to Conformal Field Theory* (Springer, 1997).

[8] D. Friedan and Z. Q. an Stephen Shenker, Phys. Rev. Lett. **52** (1984).

[9] J. L. Cardy, Nuclear Physics B **240**, 514 (1984).

[10] J. L. Cardy, Nuclear Physics B **270**, 186 (1986).

[11] J. L. Cardy, Nuclear Physics B **324**, 581 (1989).

[12] J. L. Cardy, J. Phys. A: Math. Gen **17**, L385 (1984).

[13] J. Polchinski, *String Theory Volume 1* (Cambridge University Press, 1998).

[14] J. Dai, R. G. Leigh, and J. Polchinski, Mod. Phys. Lett. A **04**, 2073 (1989).

[15] J. L. Cardy, J. Phys. A: Math Gen. **25**, L201 (1992).